

state of the button. The sequence of a click can be broken down into the following steps:

1. User moves the mouse pointer over the button
2. User presses the mouse button down
3. Program reacts by changing the image to the down state
4. User releases the mouse button
5. Program reacts by showing the up image of the button
6. Program performs some action based on the button click

Good GUIs also allow the user to click down on a button, temporarily roll off the button, changing the button to the up state, and then, with the mouse button still down, roll back over the image so the button changes back to the down image. If the user clicks down on a button but then rolls the mouse off and lifts up on the mouse button, that is not considered a click. This means the program takes action only when the user presses down and releases while the mouse is positioned over the image of a button.

Building a Button Class

The button behavior should be common and consistent for all buttons used in a GUI, so we'll build a class that takes care of the behavior details. Once we've built a simple button class, we can instantiate any number of buttons and they'll all work exactly the same way.

Let's consider what behaviors our button class must support. We'll need methods to:

- Load the images of the up and down states, then initialize any instance variables needed to track the button's state.
- Tell the button about all events that the main program has detected and check whether there are any that the button needs to react to.
- Draw the current image representing the button.